

SIDDHANTA MOHANTY

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EDUCATION

The Pennsylvania State University

Bachelor of Science in Computer Science

University Park, PA

Aug. 2022 – May 2026

- **Dean's List:** Fall 2023, Spring 2024, Fall 2025, Spring 2026 **Relevant Coursework:** Data Structures and Algorithms, Operating Systems, Artificial Intelligence, Computer Organization, Theory of Computation, Deep Learning for Computer Vision

EXPERIENCE

GenAI Intern

HCL Technologies

May 2025 – Aug. 2025

Redmond, WA

- Built a **RAG-based support automation system** using Flask, LangChain, and FAISS that analyzed production error logs, retrieved similar incidents and internal documentation, generated remediation recommendations, and sent automated Slack alerts, reducing support resolution time by **50%**
- Designed **retrieval pipelines** using FAISS and structured metadata to index internal documentation and historical incidents, improving the relevance of support recommendations
- Orchestrated **multi-agent workflows** across APIs, file systems, databases, and Confluence using **MCP** and LangChain for automated knowledge retrieval and actions

Research Assistant

Penn State University

Sep. 2025 – Dec. 2025

University Park, PA

- Designed **Reinforcement Learning** approaches to improve two-LLM code verification pipelines through iterative prompt refinement and compiler feedback
- Designed integration strategies for combining **RL** with validation tools (**Klee**, **CodeQL**) to enable automated code evaluation and self-correction

Data Analytics Intern

Elevatoz Loyalty

June 2024 – Aug. 2024

Bengaluru, India

- Analyzed datasets using **SQL** and **Python** (pandas, NumPy) and automated **ETL workflows**, reducing processing time **40%**

PROJECTS

Self-Heal System | Flask, LangChain, FAISS, PostgreSQL, GPT-3.5, Slack

[Source Code](#)

- Built an **AI self-healing pipeline** storing error context in PostgreSQL, and automating root-cause explanation and fix suggestions via LLM-driven reasoning, implemented severity-based Slack alerts reducing **MTTR 35%**

Prompt Tracer | JavaScript, Chrome Extension API, Manifest V3

[Source Code](#)

- Developed a **prompt observability Chrome extension** to track, analyze, and optimize prompts across ChatGPT, Claude, Grok, and Gemini via a real-time analytics dashboard
- Built real-time prompt evaluation pipelines to compare outputs across multiple LLMs and track performance trends, enabling iterative optimization of prompt quality

yourAlbrief | Next.js, TypeScript, Prisma, PostgreSQL, NextAuth, SendGrid, Vercel

[Demo](#) | [Source Code](#)

- Built a full-stack **AI newsletter platform** that ingests news via APIs, curates content, and delivers structured daily briefs using Next.js, Prisma, and PostgreSQL

Swipeflix | TensorFlow.js, React, Node.js, MongoDB

[Demo](#) | [Source Code](#)

- Built a personalized **movie recommender** using swipe-based feedback to model user preferences and serve real-time recommendations via a React UI, achieving **75% accuracy**

AI Transcript | FastAPI, OpenAI Whisper, LLMs, HTML/CSS/JS

[Demo](#) | [Source Code](#)

- Built a production-ready **AI pipeline** that captures audio input, performs speech-to-text transcription with **Whisper**, and uses **LLMs** to remove filler, improve flow, and generate publish-ready transcripts

TECHNICAL SKILLS

Languages: Python, C++, JavaScript, TypeScript, SQL, HTML/CSS

Frameworks/Libraries: Flask, FastAPI, Django, Node.js, REST APIs, React, LangChain

AI/ML Tools: RAG, Vector Databases (FAISS, Pinecone), LLMs, Model Context Protocol (MCP), Prompt Engineering, NumPy, Pytorch

Developer Tools: Git, Github, Docker, Postman, AWS, Google Cloud Platform, PostgreSQL, MongoDB, ROS2